CAREER AUTHENTICITY ENGINE

A Multi-Layered Machine Learning & OSINT Approach to Employment Fraud Detection

Technical Thesis & Architectural Documentation

Comprehensive Study Report

May 2026

1. EXECUTIVE SUMMARY

The **Career Authenticity Engine** is an advanced cybersecurity application designed to evaluate the legitimacy of online job postings in real-time. Operating as a Chrome Extension backed by a Python Flask server, the system employs Natural Language Processing (NLP), Artificial Neural Networks (ANN), and Open Source Intelligence (OSINT) APIs to identify recruitment fraud.

By calculating a mathematically weighted "Safety Score" based on legal registration, linguistic patterns, and infrastructure age, the engine protects users from phishing, identity theft, and employment scams. This document details the technical implementation, architectural decisions, and theoretical underpinnings of the project.

Problem Statement

> Scammers frequently clone legitimate corporate websites or post fake jobs on platforms like LinkedIn and Indeed to harvest sensitive personal data or solicit fake "processing fees." Current defensive measures fail because they rely on static blacklists that cannot keep up with dynamic fraud.

2. SYSTEM ARCHITECTURE

The system follows a modern Client-Server architecture designed for scalability and real-time response.

A. The Client (Google Chrome Extension)

- **manifest.json:** Defines permissions allowing the extension to read DOM elements.
- **content.js:** A dynamic scraper using CSS selectors to extract metadata. It uses *MutationObservers* to handle Single Page Applications (SPAs).
- **sidepanel.js:** The controller that orchestrates communication between the UI and the Flask Backend.

B. The Server (Python Flask API)

- **app.py:** The orchestration layer that standardizes company names using "Domain Peeling" and triggers concurrent verification layers.
- **verifiers.py:** Handles OSINT gathering, SSL certificate inspection, and Domain Age calculation.
- **legal_engine.py:** Manages secure communication with government endpoints.

3. THE CORE PIPELINE FLOW

When a user initiates a scan, the following sequence of high-precision events occurs:

1. **Extraction:** content.js identifies page context and extracts title, company, and description.
2. **Transmission:** Data is encrypted and sent to the Backend via HTTP POST.
3. **Legal Check (MCA):** The backend cleans the company name and queries the Indian Sandbox API to verify the Corporate Identification Number (CIN).
4. **AI Check (ANN):** The description is vectorized and processed through the Neural Network.
5. **Infrastructure Check:** The domain age is retrieved; domains under 90 days are penalized.
6. **Verdict Engine:** Results are aggregated using a weighted formula.
7. **Caching:** Results are stored in MongoDB Atlas for instant retrieval for future users.

4. ML & NATURAL LANGUAGE PROCESSING

The core intelligence of the project is a deep learning model trained on fraudulent and legitimate job datasets.

Data Preprocessing (TF-IDF Vectorization)

Neural networks require numerical input. We utilize **TF-IDF (Term Frequency-Inverse Document Frequency)** to convert text into a 5,000-dimension mathematical array.

$$TF\text{-}IDF(t, d) = TF(t, d) \times log(N / DF(t))$$

The Artificial Neural Network (MLP)

We utilize an **MLPClassifier** with three hidden layers (128, 64, and 32 neurons). This architecture identifies non-linear correlations between words like "Urgent" and "WhatsApp" in the context of job offers.

$$Output = \sigma(\Sigma\, w_i x_i + b)$$

5. EXTERNAL VERIFICATION LAYERS

To ensure 100% accuracy, the AI is supplemented by undeniable external facts.

PROVIDER	SERVICE	IMPACT ON SCORE
MCA Sandbox	Govt. Registry Check	Highest (40%)
WhoisXML	Domain Age Audit	Medium (30%)
Apify	LinkedIn Social Proof	Medium (30%)
MongoDB Atlas	Global Blacklisting	Cache/Override

6. THE WEIGHTED VERDICT ENGINE

The system prevents false alarms by using a balanced weighted algorithm:

$$Final\ Score = (MCA \times 0.40) + (ANN \times 0.30) + (Infra \times 0.30)$$

> **The "New Startup" Scenario:** A startup with a 10-day-old website (0 Infra) but a valid MCA Registration (100 Legal) and professional text (90 AI) will still result in a 67% (Safe/Caution) score, proving the system is context-aware.

7. THEORETICAL VIVA & EXAMINATION PREP

Potential questions and technical answers for project defense:

Q: Why use an ANN instead of a Keyword Scanner?

Keyword scanners are easily bypassed by character obfuscation. ANNs analyze contextual density and syntactical structure, making evasion significantly harder.

Q: How did you solve the MongoDB SSL errors on Windows?

Windows often fails to locate root certificates. We solved this by using the 'certifi' library and passing the CA bundle directly into the MongoClient.

Q: What is "Domain Peeling"?

It is an algorithm we implemented to strip away Applicant Tracking System suffixes (like .skima.ai) to find the true employer's domain for WHOIS checks.

8. LIMITATIONS & FUTURE SCOPE

Current Limitations

- **Language Barrier:** Optimized for English; regional language support is pending.
- **API Latency:** Real-time scans are dependent on external API response times.
- **Zero-Day Novelty:** Extremely novel scam techniques may result in false negatives.

Future Roadmap

- **BERT Integration:** Moving from TF-IDF to BERT embeddings for deeper context.
- **Blockchain Ledger:** Decentralized logging of verified company identities.
- **Image OCR:** Scanning job posters and images for hidden contact numbers.

9. CONCLUSION

The **Career Authenticity Engine** represents a significant step forward in recruitment security. By moving beyond text analysis and into infrastructural and legal verification, the project provides a robust shield for the modern job seeker.

$$Multi\text{-}Layer\ Defense = Security^{\infty}$$